

MOHAMAD ARIYANTO

Automation Engineering · Fresh Graduate

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Surabaya, Indonesia

PROFESSIONAL SUMMARY

Final-year Automation Engineering student at Surabaya State Shipbuilding Polytechnic (PPNS) with hands-on experience in industrial automation, PLC programming, IoT integration, and embedded systems. Proven ability to contribute to real-world projects through an internship at PT. Paksi Jaya Santosa and ongoing research at BRIN. Passionate about driving digital transformation in shipbuilding and manufacturing industries through smart automation and real-time monitoring systems.

EDUCATION

Politeknik Perkapalan Negeri Surabaya (PPNS)

Applied Bachelor's Degree – Automation Engineering | GPA: 3.36 / 4.00 2022 – 2026 (Expected)

Relevant Coursework: Industrial Automation, PLC Programming, Embedded Systems, IoT Integration, Electrical Control System, HMI & Monitoring Systems

PROFESSIONAL EXPERIENCE

BADAN RISET DAN INOVASI NASIONAL (BRIN) | Surabaya, Indonesia

Research Internship – Final Project Research

March 2026 – July 2026

- Conducted research and development in automation and embedded systems for final project implementation.
- Assisted in system testing, data acquisition, and research documentation.
- Developed prototype systems involving IoT integration and real-time monitoring technology.
- Collaborated with researchers and technical teams in research-based engineering activities.

PT. PAKSI JAYA SANTOSA | Jakarta, Indonesia

Internship – Automation & PLC Engineer

April – November 2025

- Contributed to automation control projects for Ammolift and Foodlift systems at PT. Noahtu Shipyard.
- Assisted in Siemens PLC programming, troubleshooting, and control panel integration.
- Participated in electrical control system testing and commissioning for industrial lifting systems.
- Supported system monitoring, wiring inspection, and automation implementation in shipbuilding environments.
- Collaborated with engineering teams to ensure automation systems operated safely and efficiently.

TECHNICAL & PROFESSIONAL SKILLS

- PLC Programming & Troubleshooting (Siemens, Omron)
- IoT Integration (ESP32, ESP8266, Arduino, Blynk)
- Embedded Systems & Microcontroller Programming
- HMI Design, Debugging & SCADA Systems
- Programming: C++, Python
- Electronic Design (Eagle, Proteus, Fritzing)
- Sensor & Actuator Calibration (Temperature, Pressure, Proximity)
- Web-Based Monitoring Application Development
- Machine Vision & Image Processing
- Data Acquisition & Research Documentation
- Analytical Thinking & Problem Solving
- Scientific & Technical Writing
- Team Collaboration & Communication
- Adaptability in Industrial Environments

PROJECTS

Inspection and Test Plan (ITP) Monitoring System (2026)

- Developed a web-based ITP monitoring system for shipbuilding production with workflow management, database integration, and multi-user access control.

Aluminum Plate Defect Detection Using Machine Vision (2025)

- Developed an automated defect detection system using machine vision techniques for quality control of aluminum plates in manufacturing.

Prototype Anti-Heeling System with Ballast for Ship Stability (ESP32-Based) (2025)

- Designed an ESP32-based anti-heeling system using ballast control to maintain ship stability during docking, with real-time sensor integration and automated balance control.

Automated Conveyor Sorting System Using Omron PLC (2025)

- Designed a sensor-based automated sorting system using Omron PLC for efficient industrial material handling and product classification.

ESP32-Based Autonomous Line Follower Robot (2025)

- Built an autonomous line follower robot with a custom sensor system, integrating embedded programming, sensor calibration, and motor control for stable navigation.

Automatic Fire Suppression System Using PLC CX-Programmer (2024)

- Implemented an Omron PLC-based fire suppression system with fire detection sensors, alarm integration, and automated extinguishing control for industrial safety.

ORGANIZATIONAL EXPERIENCE

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| ▸ Minister, Research & Science Ministry – BEM PPNS | <i>2024 – 2025</i> |
| ▸ Expert Staff, Research & Science Ministry – BEM PPNS | <i>2023 – 2024</i> |
| ▸ Expert Staff, Home Affairs Department – BISHIPS PPNS | <i>2023 – 2024</i> |
| ▸ Apprentice Staff, Home Affairs Department – BISHIPS PPNS | <i>2022 – 2023</i> |

CERTIFICATIONS

2nd Conference On Automation Engineering & Its Application (CAEA)

- Participated in a conference covering industrial automation, control systems, and engineering applications.
- Expanded knowledge of emerging automation technologies and industry practices.

Test of English for International Communication (TOEIC)

- Certification of English language skills in the context of international communication, especially for professional and technical work needs. With a TOEIC score of 525 with listening 210 and reading 315.